Factory Reallocation and Shipping Optimization for Government Stakeholders

Executive summary | Independent educational analysis | 10 October 2026

Decision for stakeholder review

Support verification of logistics data and a narrowly scoped feasibility review before considering factory reassignment. The public educational sample identifies where further investigation is most useful, but it does not establish an operational business case, realized public benefits or a case for public investment.

What the sample establishes

The 2021-2024 sample has 10,194 records and 38,654 units, with sales of 141,783.63 and source gross profit of 93,442.80. Monetary units are unspecified; gross profit excludes freight and other operating costs. Chocolate represents 96.43% of demand. Geographic screening identifies five persistently distant product-region flows, comprising two Pacific and three Atlantic chocolate flows.

There are **zero validated historical lead-time records**. Every source interval is 2,000-2,011 days. Delivery times cannot be measured because delivery dates are absent. These failures prevent a verified performance or reassignment claim.

What the planning exercise suggests

At equal speed and profit priority, the assumption-based simulator flags Nutty Crunch Surprise, Fudge Mallows, Scrumdiddlyumptious, Milk Chocolate and Triple Dazzle Caramel for investigation at Secret Factory. These are hypothetical product-factory alternatives; none has historical outcome evidence or verified production constraints.

Planning indicators are not observed service improvements or outcome probabilities

Planning indicator	Default result
Modeled dispatch allowance reduction	4.06%
Nonnegative joint contribution cases	54 of 81 cases or 66.67%
Evidence readiness and verified coverage	0 of 100 and 0%
Products proposed for investigation	5 of 15 or 33.33%

The base modeled contribution change is +977.68 source monetary units over the mapped sample replay. Tested joint changes range from -4,090.69 to +4,066.00. No alternate assignment protects contribution in every sensitivity case. These figures are not annual savings, statistical confidence bounds or monetary benefits to government.

Proposed next steps

1 Establish reliable milestones. Ask the data owner to verify order and ship dates, observed origin factories and delivery dates. Preserve the source records and document verified corrections. Retain Canadian demand in business totals and obtain destination coordinates for geographic comparisons.

2 Review the most exposed flows. Collect actual freight routes and tariffs for Milk Chocolate and Triple Dazzle Caramel in Pacific, and Nutty Crunch Surprise, Fudge Mallows and Scrumdiddlyumptious in Atlantic. The geographic screen highlights distance exposure; it does not show congestion, emissions or service failure.

3 Verify production feasibility. Obtain documented product compatibility, additional factory capacity over an agreed horizon, manufacturing differences, setup costs and transition requirements. The default proposals send 37,275 whole-product sample units to Secret Factory, compared with 884 units statically assigned there in the sample. Neither figure measures physical capacity; concentration makes verification essential.

4 Evaluate a limited trial if evidence supports it. Define eligible products and service requirements before any operational change. Compare observed dispatch and delivery performance and fully scoped costs with retained current assignments. An analyst should maintain the data/assumption audit, operating owners should validate capability and costs, and stakeholder reviewers should assess the measured evidence. These are proposed responsibilities, not existing commitments.

Conditions for a credible decision

A reassignment proposal should include validated shipment milestones, observed origin records, product-factory outcome evidence, capacity and capability checks, a complete transition-cost comparison, and independent evaluation on later observations. Report losses and service deterioration alongside gains. A positive planning score or clean software test does not establish a feasible or profitable move.

The sample contains no basis for estimating employment effects, emissions reductions, tax receipts, industry-wide congestion or public return on expenditure. Any such assessment requires separate evidence. No budget, plant-relocation incentive or infrastructure commitment is recommended from the current figures.

Delivery and evidence status

The completed project includes an auditable research paper, Streamlit simulator and risk panel, downloadable scenario results, source-data attribution and reproducible code. An independently extracted package reproduced its saved KPIs and passed 24 software tests. Account publication remains pending. This is an educational submission, not an official company study or government policy evaluation.

Evidence: datasets/cleaning_validation_summary.json; reports/eda/current_performance_report.md; reports/route_analysis/route_investigation_report.md; reports/recommendations/kpis.json and recommendation_report.md. The accompanying research paper supplies the full methods, metrics and limitations. Order sample: Maven Analytics US Candy Distributor; ZIP geography: SimpleMaps historical Basic snapshot, CC BY 4.0. All monetary figures use unspecified source units.